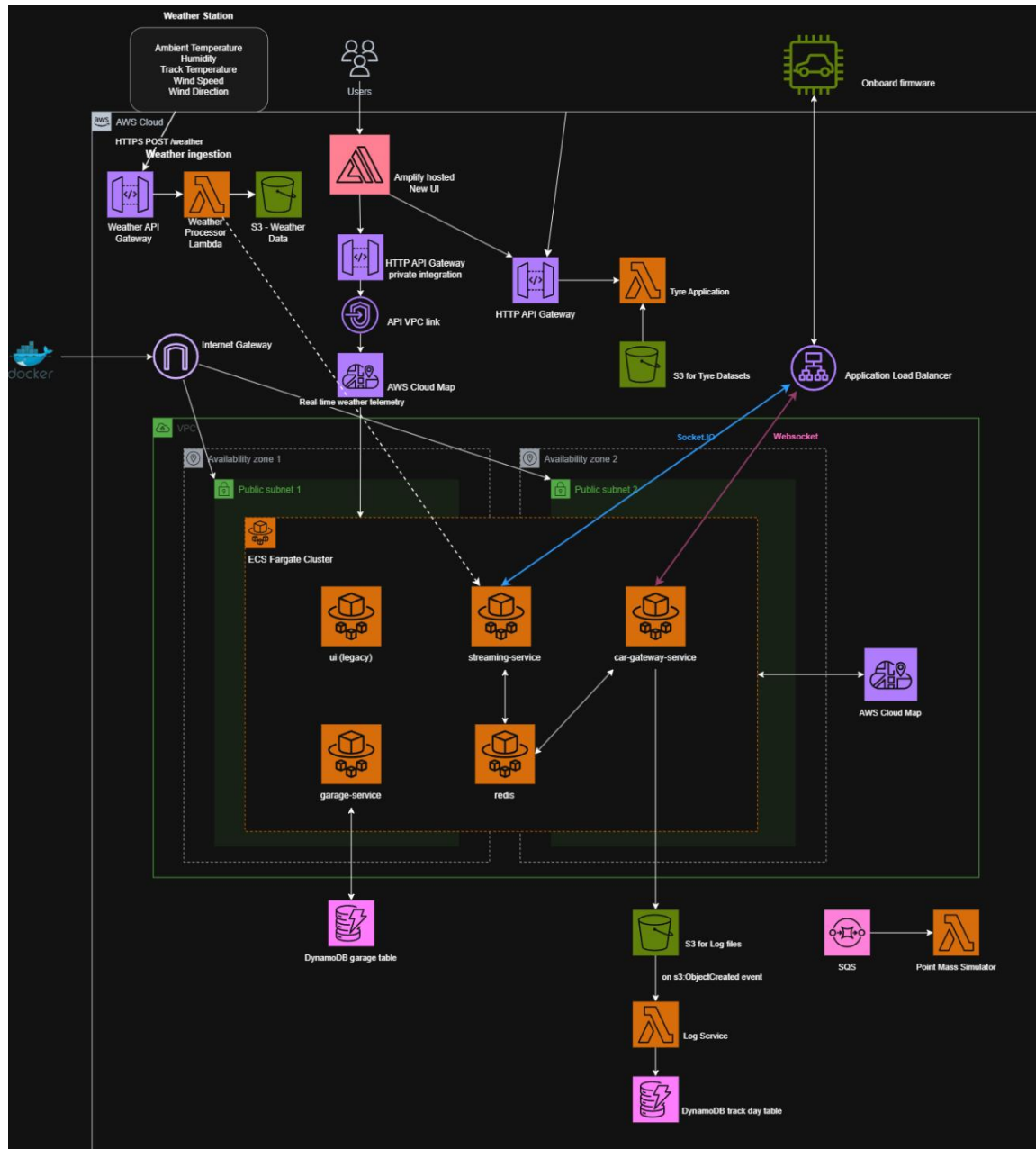


Vehicle Analytics Cloud Technical Assessment

Local Weather Station Integration

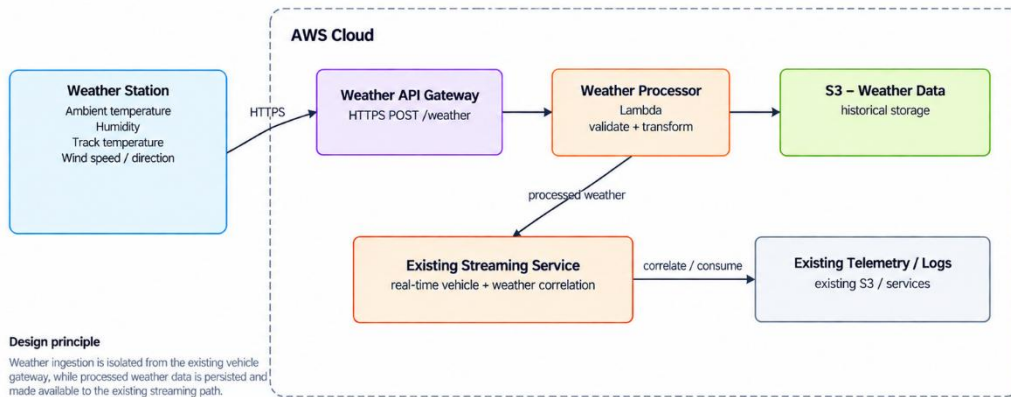
1. Architecture Overview



The system adds a weather data collection feature to the Vehicle Analytics platform. Weather data is sent to an API Gateway, processed by a Lambda function, and stored in S3. It can also be combined with vehicle telemetry data for real-time analysis.

Updated Weather Station Architecture

Conceptual AWS architecture for integrating local weather data into Vehicle Analytics



2. Data Ingestion

The weather station is located near the track and collects weather data. It sends the data to Amazon API Gateway through HTTPS. A **POST /weather** endpoint is used to receive the weather measurements.

Ambient temperature

Humidity

Track temperature

Wind speed

Wind direction

Timestamp

3. Data Processing and Storage

The Weather Processor Lambda checks the incoming weather data and removes invalid records. Valid data is then converted into a standard format and stored in S3 for future analysis.

4. Integration with Existing Telemetry

Processed weather data can also be sent to the existing streaming service. Since both weather data and vehicle telemetry data have timestamps, engineers can compare them to see how weather conditions affect vehicle performance.

5. Security

Communication between the weather station and API Gateway uses HTTPS. Authentication should be applied to the weather endpoint. Lambda should use a least-privilege IAM role, while the S3 bucket should block public access and use encryption at rest.

6. Reliability and Error Handling

Invalid weather data is rejected and not stored. AWS logging and monitoring services are used to track API Gateway and Lambda errors. If more buffering is needed in the future, an SQS queue can be added between data ingestion and processing.

7. Scalability and Cost

API Gateway and Lambda allow the system to process weather data without running a server all the time. S3 is used to store historical weather data safely. This design is simple and cost-effective for a small number of weather stations.

8. Infrastructure as Code

The infrastructure can be deployed using Terraform or AWS CDK. The IaC code creates the API Gateway, Lambda function, IAM permissions, S3 bucket, monitoring services, and the connection to the existing streaming service. The infrastructure code should be stored in version control and separated into different environments, such as development and production.

9. Design Trade-offs

Decision	Benefit	Trade-off
API Gateway	Simple HTTPS API	Requires authentication

Lambda	No always-on server	Runtime limitations
S3	Durable and inexpensive storage	Not designed for live state
Separate weather path	Keeps vehicle system isolated	Adds another service path
Streaming integration	Enables real-time correlation	Requires stable data contract